

EMBEDDED HACKING

Reverse Engineering the Raspberry Pi Pico 2

Operation Dark Vector: Classified Intelligence Briefing

ARM Cortex-M33 • GDB • Ghidra

October 1, 2026

For educational and defensive security research only

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|                TOP SECRET // NOFORN                |
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| OPERATION DARK VECTOR                               |
| CLASSIFIED BRIEFING: LIVE CTF 0x01                  |
| NATIONAL SECURITY AGENCY / GMU RHET                  |
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1. The Answer

INT. NSA EXPLOITATION CELL. 05:03.

The analysts have been awake for nineteen hours. The Director walks in with a cup of coffee and does not sit down.

DIRECTOR: Where is it going?

ANALYST: Buddy says it knows.

DIRECTOR: Buddy.

ANALYST: The model. The one we built for exactly this. Give it the image, ask it the question.

DIRECTOR: And?

ANALYST: Nine seconds.

She turns the screen around. This is what came back.

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| BUDDY // CLASSIFIED EXPLOITATION ASSIST // CONF: 0.97 |
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| DESTINATION : +38.840280 -77.428890                  |
| CONFIDENCE  : 0.97                                   |
| PATCH       : verified-by-construction. Ready to flash. |
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DIRECTOR: That is the whole answer.

ANALYST: That is the whole answer.

DIRECTOR: Is it that simple?

ANALYST: It is never that simple.

DIRECTOR: Then why does it look that simple?

ANALYST: Because Buddy is very good at making things look simple.

2. What Buddy Is

Buddy is not a person. It is a model. It has read more assembly than every engineer who has ever lived, and it never sleeps and never doubts. You ask it a question, and it answers in the exact shape of an answer, with a confidence that is hard to argue with.

That confidence is the problem. It is not the same thing as being right.

Buddy has never seen this drone. It has never stood at this bench or watched this board power on. It has a picture of the firmware, and it has made a very good guess. A very good guess is still a guess.

3. The Operation

At 04:17 the woods outside Centreville, Virginia were black and the machine above them was silent. It was a one-way drone, built by a state program we will not name here, meant to fly to a target, do its work, and never come home. No radio, no hand on the stick. A breadboard-ugly brain counting down a heading it was born with.

The wind won. The battery died. It came down through the branches and stayed there, about a mile from where it started.

Then it started talking. A recovery beacon on 915 MHz, repeating a coordinate into the night. NSA sensors heard it, and a recovery team took the airframe intact. The flight computer was a bare-metal RP2350 that was never supposed to be opened, and the one thing it was still willing to say out loud.

CyberCom imaged the flash and did what everyone does now. They handed it to the machine.

4. The Mandate

The Director does not trust the answer. Not because Buddy is bad, but because the answer is too clean. Nine seconds for a question that should have taken a night.

So the order is simple, and it is the only order that matters.

Prove it. Prove Buddy correct, or prove it wrong with evidence that no one can argue with. A machine's verdict does not move up the chain without a human signature.

5. The Challenge

You have the image. You have the board. You have the tools and the time Buddy did not need.

Buddy has given you its answer, and it is confident.

Is it that simple?

Go find out.